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BSIT-4A

Enterprise Server Deployment and Operating System Installation

Installation Guide

1. Project Overview

This documentation outlines the end-to-end setup of Ubuntu Server and Windows Server environments within VirtualBox, simulating a Junior System Administrator role. Key tasks included building virtual machines, configuring core network settings, securing remote access via SSH, and updating system packages to support future enterprise services like database hosting, web applications, and file sharing. The report concludes with a technical comparison of BIOS/UEFI firmware and leading enterprise server OS platforms.

2. Objectives

The objectives of this project were to:

- Install Ubuntu Server in a virtual machine.
- Configure the hostname, user account, storage, network, and keyboard settings.
- Install and verify the OpenSSH Server service.
- Verify the server hostname, IP address, and internet connectivity.
- Update the Ubuntu Server operating system.
- Install and configure Windows Server Evaluation.
- Compare BIOS and UEFI firmware.
- Illustrate the Ubuntu boot process.
- Compare Windows Server, Ubuntu Server, and Rocky Linux.
- Produce clear technical documentation for future administrators.

3. Software Requirements

The following software and installation media were used:

- Oracle VirtualBox 7.2
- Ubuntu Server 24.04.4 LTS ISO
- Windows Server 2022 Evaluation ISO

- Microsoft Word
- Draw.io or an equivalent diagramming tool

4. Ubuntu Server Virtual Machine Specifications

Component	Configuration
Virtual Machine Name	Ubuntu-Server-Week03
Operating System	Ubuntu 64-bit
Memory	4096 MB
Processors	2 virtual processors
Virtual Storage	40 GB VDI
Network	NAT
Installation Media	Ubuntu Server 24.04.4 LTS AMD64 ISO
Hostname	server01
Administrative Username	nico

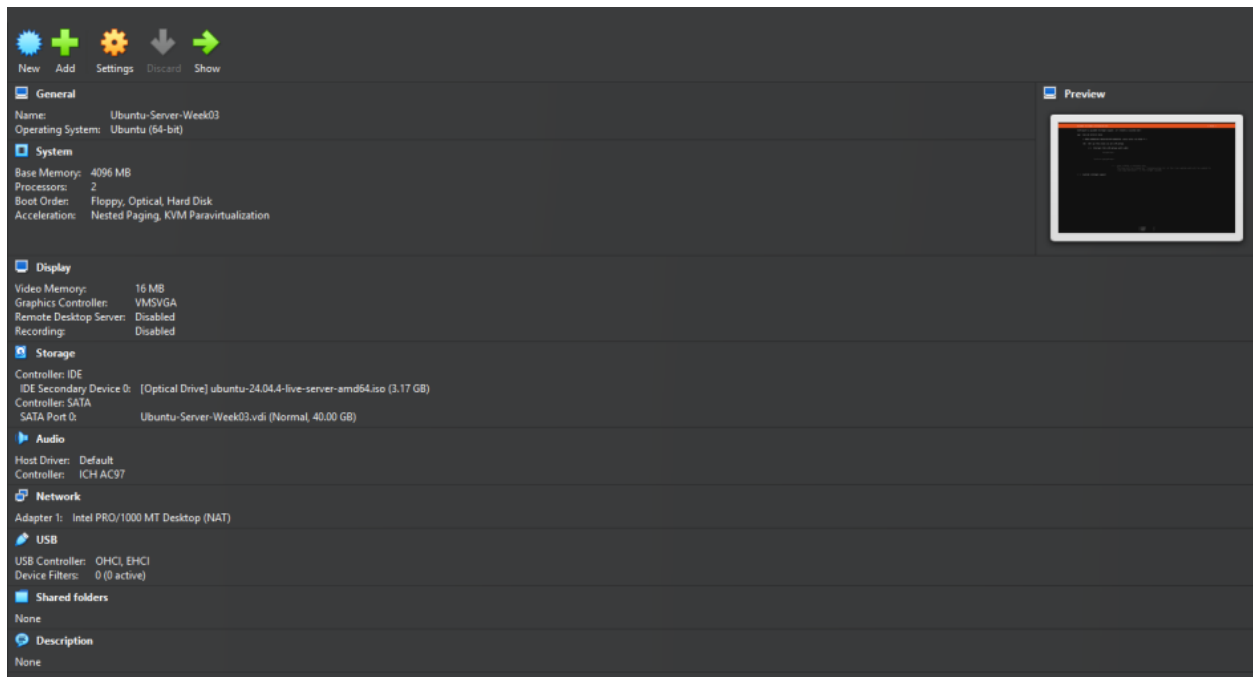


Figure 1. Oracle VirtualBox specifications for the Ubuntu Server virtual machine.

5. Ubuntu Server Installation Procedure

5.1 Starting the Installer

The Ubuntu Server ISO was mounted to the virtual optical drive prior to powering on the VM. From the boot screen, the **Try or Install Ubuntu Server** option was selected, with

English set as the system language and English (US) chosen for the keyboard layout. The installer's update prompt was intentionally bypassed to allow package updates to be executed and documented manually during the post-installation verification phase.

5.2 Network Configuration

Network connectivity was established using VirtualBox's default Network Address Translation (NAT) mode, with the Ubuntu installer automatically receiving an IP address via DHCP. Because the VM utilized a single network interface, network bonding was unnecessary. The proxy configuration was skipped as none was required for the environment, and the default Ubuntu archive mirror was kept for package downloads.

5.3 Storage Configuration

Guided storage configuration was selected using the entire 40 GB virtual disk. Logical Volume Management was enabled using the default settings. Disk encryption was not enabled because it was not required for the laboratory activity. Ubuntu created a BIOS boot partition, an ext4 /boot partition, and an LVM logical volume formatted as ext4 and mounted at /.

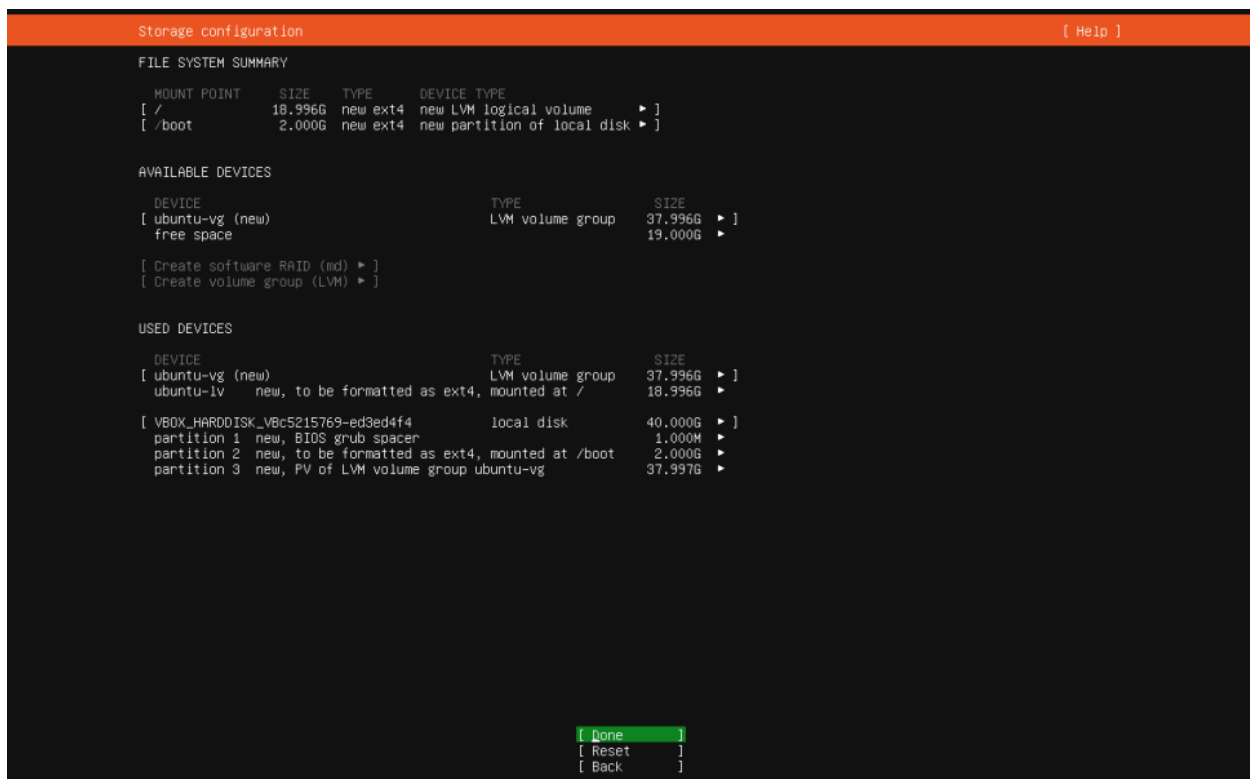
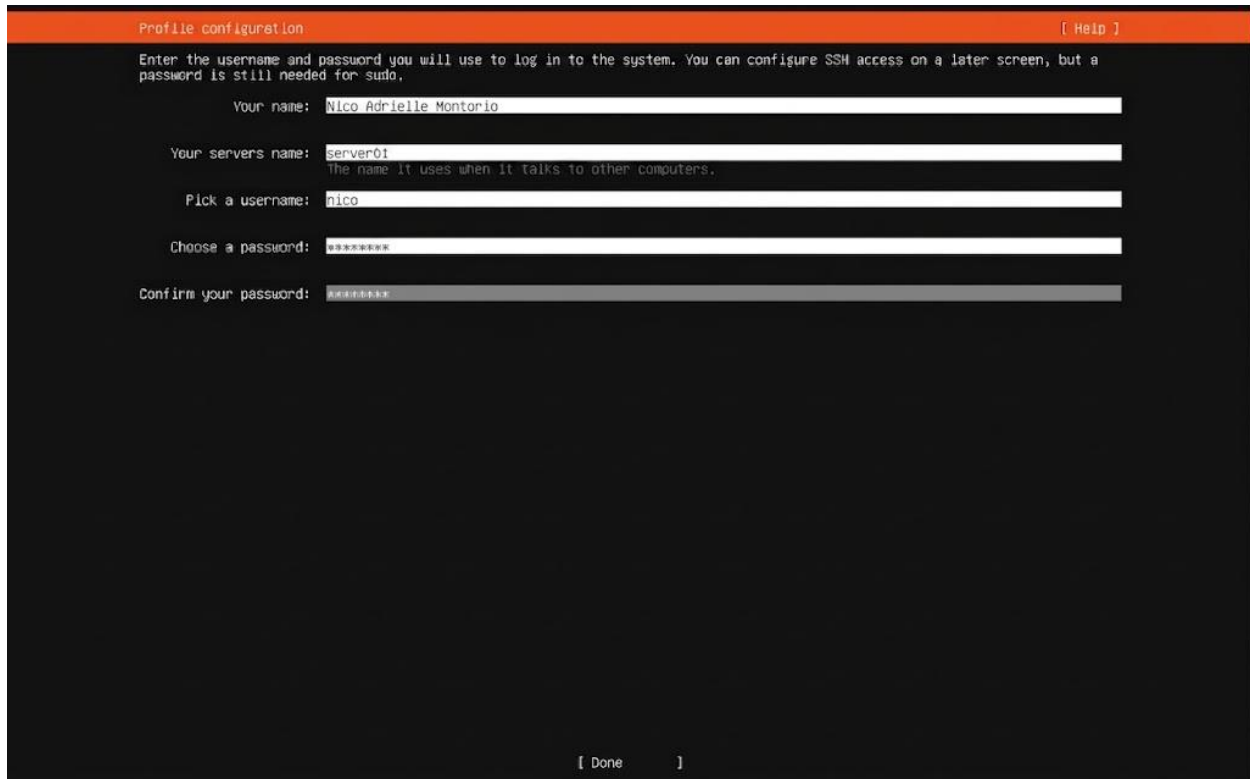


Figure 2. Ubuntu Server storage configuration showing the partitions, LVM volume, and ext4 file system

5.4 User and Hostname Configuration

During profile setup, a non-root administrative account was created using nico as the username alongside the system hostname server01. A secure password was defined for both local authentication and elevated privileges via sudo.



The screenshot shows the 'Profile configuration' window with an orange header bar. The header contains the title 'Profile configuration' on the left and a '[Help]' link on the right. Below the header, a message reads: 'Enter the username and password you will use to log in to the system. You can configure SSM access on a later screen, but a password is still needed for sudo.' The form contains five input fields: 'Your name:' with the value 'Nico Adrielle Montorio', 'Your servers name:' with the value 'server01' and a subtext 'The name it uses when it talks to other computers.', 'Pick a username:' with the value 'nico', 'Choose a password:' with masked characters '*****', and 'Confirm your password:' with masked characters '*****'. At the bottom center, there is a '[Done]' button.

Figure 3. Ubuntu Server profile configuration showing the server hostname and administrative user.

5.5 Additional Configuration

Ubuntu Pro was skipped because it was not required for the activity. No featured server snaps were installed because the instructions required a clean server installation without unnecessary additional packages.

The installer then copied and configured the required operating-system files

```
Installing system [ Help ]

subiquity/Ad/apply_autoinstall_config:
subiquity/Late/apply_autoinstall_config:
configuring apt
curtin command in-target
installing system
executing curtin install initial step
executing curtin install partitioning step
curtin command install
  configuring storage
    running 'curtin block-meta simple'
  curtin command block-meta
    removing previous storage devices
    configuring disk: disk-sda
    configuring partition: partition-0
    configuring partition: partition-1
    configuring format: format-0
    configuring partition: partition-2
    configuring lvm_volgroup: lvm_volgroup-0
    configuring lvm_partition: lvm_partition-0
    configuring format: format-1
    configuring mount: mount-1
    configuring mount: mount-0
  executing curtin install extract step
  curtin command install
    writing install sources to disk
    running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmpg4s3y12n/mount
  configuring keyboard
  curtin command in-target
  executing curtin install curthooks step
  curtin command install
    configuring installed system
    running 'curtin curthooks'
  curtin command curthooks
    configuring apt
    installing missing packages
    installing packages on target system: ['grub-pc']
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel /
```

[View full log]

Activate Wi-Fi
Go to Settings

Figure 4. Ubuntu Server installation progress

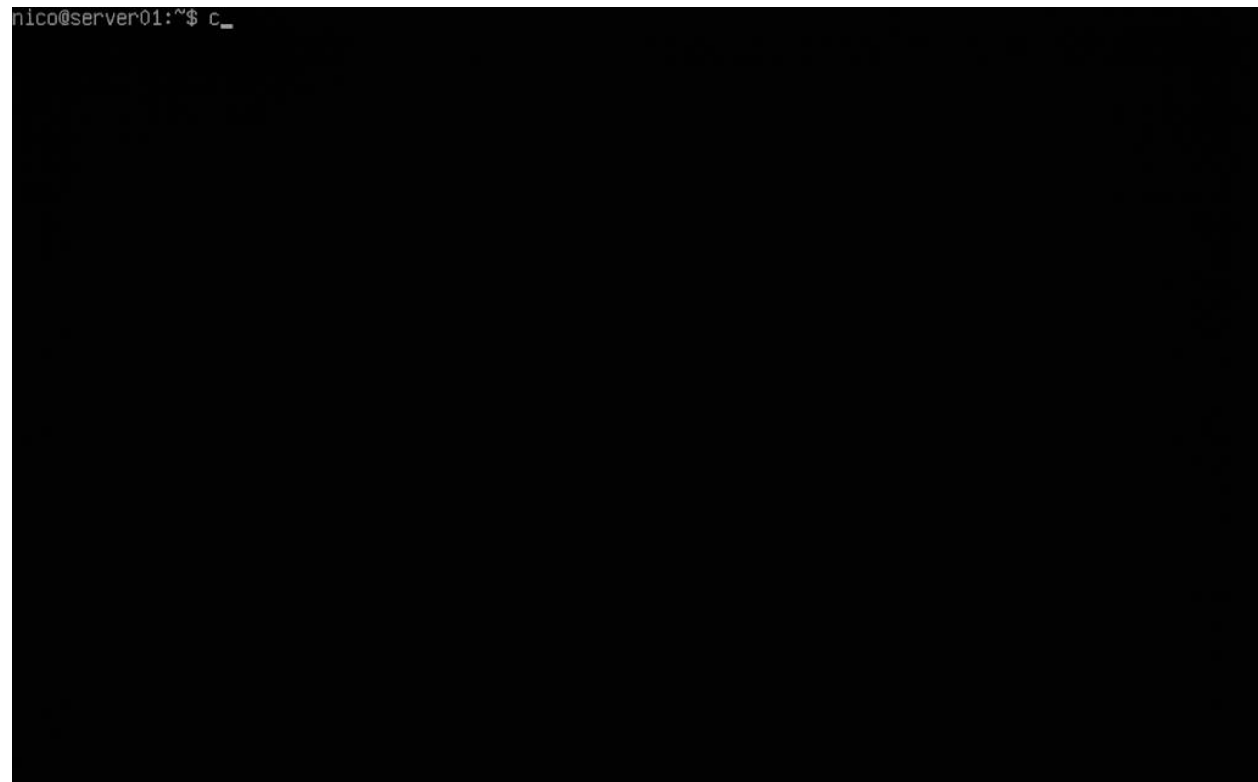
5.6 Completing the Installation

After completing the installation, the ISO image was unmounted from the virtual optical drive. The VM was then rebooted to boot directly into the newly installed operating system on the virtual hard disk.

6. Ubuntu Server Verification

6.1 Successful Login

After rebooting, the administrative user logged in using the account created during installation. The prompt `nico@server01` confirmed the username and configured server hostname.

A terminal window with a black background. The prompt 'nico@server01:~\$' is visible at the top left, followed by a cursor and the letter 'c'.

```
nico@server01:~$ c_
```

Figure 5. Successful login to Ubuntu Server using the non-root administrative account

6.2 Hostname Verification

The following command was used:

- Hostname

The command returned:

- Server01

This confirmed that the hostname was configured correctly.

```
nico@server01:~$ hostname
server01
nico@server01:~$ _
```

Figure 6. Verification of the Ubuntu Server hostname

6.3 IP Address Verification

The assigned network address was checked using:

- ip addr

The enp0s3 network interface displayed the IPv4 address 10.0.2.15/24. The interface also showed an active state, confirming that the VirtualBox network adapter was functioning.

```

lem@server01:~$ hostname
server01
lem@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c9:a0:79 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86096sec preferred_lft 86096sec
    inet6 fd00::a00:27ff:fec9:a079/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86098sec preferred_lft 14098sec
    inet6 fe80::a00:27ff:fec9:a079/64 scope link
        valid_lft forever preferred_lft forever
lem@server01:~$

```

Figure 7. Network interface and DHCP-assigned IP address of Ubuntu Server

6.4 Internet Connectivity Test

Internet and Domain Name System connectivity were tested using:

- `ping -c 4 google.com`

The server successfully received four responses, confirming that it could resolve the domain name and communicate through the internet.

```

nico@server01:~$ hostname
server01
nico@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c9:a0:79 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic ens3
        valid_lft 86096sec preferred_lft 86096sec
    inet6 fd00::a00:27ff:fec9:a079/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86096sec preferred_lft 14096sec
    inet6 fd00::a00:27ff:fec9:a079/64 scope link
        valid_lft forever preferred_lft forever
nico@server01:~$ ping -c 4 google.com
PING google.com (172.217.27.78) 56(84) bytes of data:
64 bytes from ncmnlh-aj-in-f14.1e100.net (172.217.27.78): icmp_seq=1 ttl=255 time=10.1 ms
64 bytes from ncmnlh-aj-in-f14.1e100.net (172.217.27.78): icmp_seq=2 ttl=255 time=8.78 ms
64 bytes from ncmnlh-aj-in-f14.1e100.net (172.217.27.78): icmp_seq=3 ttl=255 time=12.2 ms
64 bytes from ncmnlh-aj-in-f14.1e100.net (172.217.27.78): icmp_seq=4 ttl=255 time=12.7 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3010ms
rtt min/avg/max/mdev = 8.781/10.926/12.630/1.576 ms
nico@server01:~$ _

```

Figure 8. Successful internet connectivity test using the ping command.

6.5 System Update

The server's package information and installed packages were updated using:

- sudo apt update
- sudo apt upgrade -y

The first command retrieved the latest package information from the configured Ubuntu repositories. The second command installed available package and security updates.

```

update-rc.d: warning: start and stop actions are no longer supported; falling back to defaults
Setting up libfupd2:amd64 (1.9.34-ubuntu124.04.1) ...
Setting up libfupd3:amd64 (2.0.20-ubuntu124.04.2) ...
Setting up software-properties-common (0.99.49.4) ...
Setting up multipath-tools (0.9.4-ubuntu08.2) ...
Setting up netplan.io (1.1.2-0ubuntu124.04.2) ...
Setting up plymouth-theme-ubuntu-text (24.04.00-ubuntu7.2) ...
update-initramfs: deferring update (trigger activated)
Setting up fupdp (2.0.20-ubuntu124.04.2) ...
Installing new version of config file /etc/fupdp/bios-settings.d/README.md ...
Installing new version of config file /etc/fupdp/remotes.d/lvfs-testing.conf ...
Installing new version of config file /etc/fupdp/remotes.d/lvfs.conf ...
Installing new version of config file /etc/fupdp/remotes.d/vendor-directory.conf ...
Installing new version of config file /etc/grub.d/35_fupdp ...
Installing new version of config file /etc/update-notif.d/05-fupdp ...
fupdp-refresh.service is a disabled or a static unit not running, not starting it.
fupdp.service is a disabled or a static unit not running, not starting it.
Setting up cloud-init (26.1-0ubuntu124.04.1) ...
Installing new version of config file /etc/cloud/templates/chrooty.conf.freebsd.tpl ...
Setting up apport-core-dump-handler (2.28.3-0ubuntu1) ...
Setting up apport (2.28.3-0ubuntu1) ...
apport-autoreport.service is a disabled or a static unit not running, not starting it.
Processing triggers for dbus (1.14.16-0ubuntu4.1) ...
Processing triggers for udev (255.4-1ubuntu0.17) ...
Processing triggers for install-info (7.1-3build2) ...
Processing triggers for libc-bin (2.35-0ubuntu8.8) ...
Processing triggers for rsyslog (8.2312.0-3ubuntu9.3) ...
Processing triggers for nan-dp (2.12.0-4build2) ...
Processing triggers for initramfs-tools (0.142ubuntu25.0) ...
update-initramfs: Generating /boot/initrd.img-6.8.0-137-generic
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
/etc/needrestart/restart.d/systemd-manager
systemctl restart packagekit.service

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service

No containers need to be restarted.

User sessions running outdated binaries:
nico @ user manager service: systemd[1197]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
nico@server01:~$

```

Figure 9. Ubuntu Server package and security update process.

6.6 OpenSSH Server Verification

OpenSSH Server was installed using:

- `sudo apt install openssh-server -y`

The service was started and verified using:

- The service was started and verified using:
- `sudo service ssh status`

The status displayed active (running), confirming that the server was ready for secure remote administration.

```

nico@server01:~$ sudo service ssh status
• ssh.service - OpenBSD Secure Shell server
  Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
  Active: active (running) since Wed 2025-08-14 17:25:07 UTC; 42s ago
  TriggeredBy: • ssh.socket
  Docs: man:sshd(8)
        man:sshd_config(5)
  Process: 13983 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Main PID: 13984 (sshd)
  Tasks: 1 (limit: 4601)
  Memory: 1.2M (peak: 1.3M)
  CPU: 25ms
  CGroup: /system.slice/ssh.service
          └─13984 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 14 17:25:07 server01 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 14 17:25:07 server01 sshd[13984]: Server listening on 0.0.0.0 port 22.
Aug 14 17:25:07 server01 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Aug 14 17:25:07 server01 sshd[13984]: Server listening on :: port 22.
nico@server01:~$

```

Figure 10. OpenSSH Server service showing an active and running status.

7. BIOS and UEFI Comparison

Category	BIOS	UEFI
Definition	Traditional firmware used to initialize hardware and begin loading an operating system.	Modern firmware interface connecting the operating system with the platform firmware.
Boot Process	Traditional firmware used to initialize hardware and begin loading an operating system.	Loads the operating-system bootloader from the EFI System Partition.
Disk Support	Traditional firmware used to initialize hardware and begin loading an operating system.	GPT supports disks significantly larger than 2 TB.
Partition Style	Master Boot Record.	GUID Partition Table.
Security	Provides limited boot verification.	Supports Secure Boot and signed boot components.
Speed	Generally slower during initialization.	Can provide faster startup and hardware initialization.
Advantages	Simple and compatible with older systems.	Supports large disks, additional partitions, modern security, and flexible firmware management.

Disadvantages	Limited storage, partitions, security, and extensibility.	Can be more complicated and may not support older operating systems.
Modern Usage	Mainly used by legacy computers.	Standard firmware for modern computers and servers.

Due to its superior suitability for contemporary storage, UEFI has largely supplanted BIOS. Hardware, operating systems, and security specifications. It makes use of GPT to enable larger more partitions and storage devices than a conventional MBR. UEFI Secure Boot is capable of additionally check boot components before letting them run, lowering the possibility of unapproved software that runs at startup. Because of these features, UEFI is more. Secure and scalable for contemporary enterprise deployments

The complete comparison and one-page discussion are provided separately in BIOS_vs_UEFI.pdf.

8. Ubuntu Boot Process

The Ubuntu Server boot process consists of eight major stages:

1. Power On
2. BIOS/UEFI Initialization
3. Boot Device Detection
4. Boot Loader (GRUB)
5. Linux Kernel
6. Init/systemd
7. Services Start
8. Login Prompt

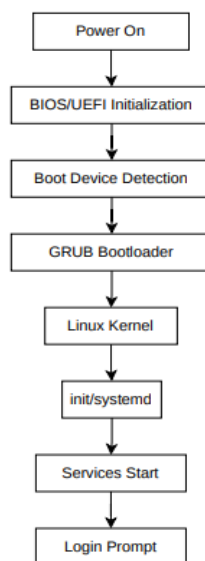


Figure 11. Ubuntu Server boot-process flowchart.

During startup, the firmware initializes the virtual hardware and detects the bootable storage device. GRUB then loads the Linux kernel. The kernel initializes memory, processors, and device drivers before starting systemd. Systemd starts networking, SSH, and other required services before displaying the login prompt.

9.Windows Server Installation

9.1 Virtual Machine Configuration

A separate virtual machine was created for Windows Server to avoid modifying the completed Ubuntu Server installation.

Component	Configuration
Virtual Machine Name	Windows-ServerWeek03
Operating System	Windows Server 2022 Evaluation
Memory	4096 MB
Processors	2 virtual processors
Virtual Storage	50 GB
Network	NAT
Computer Name	SERVER02

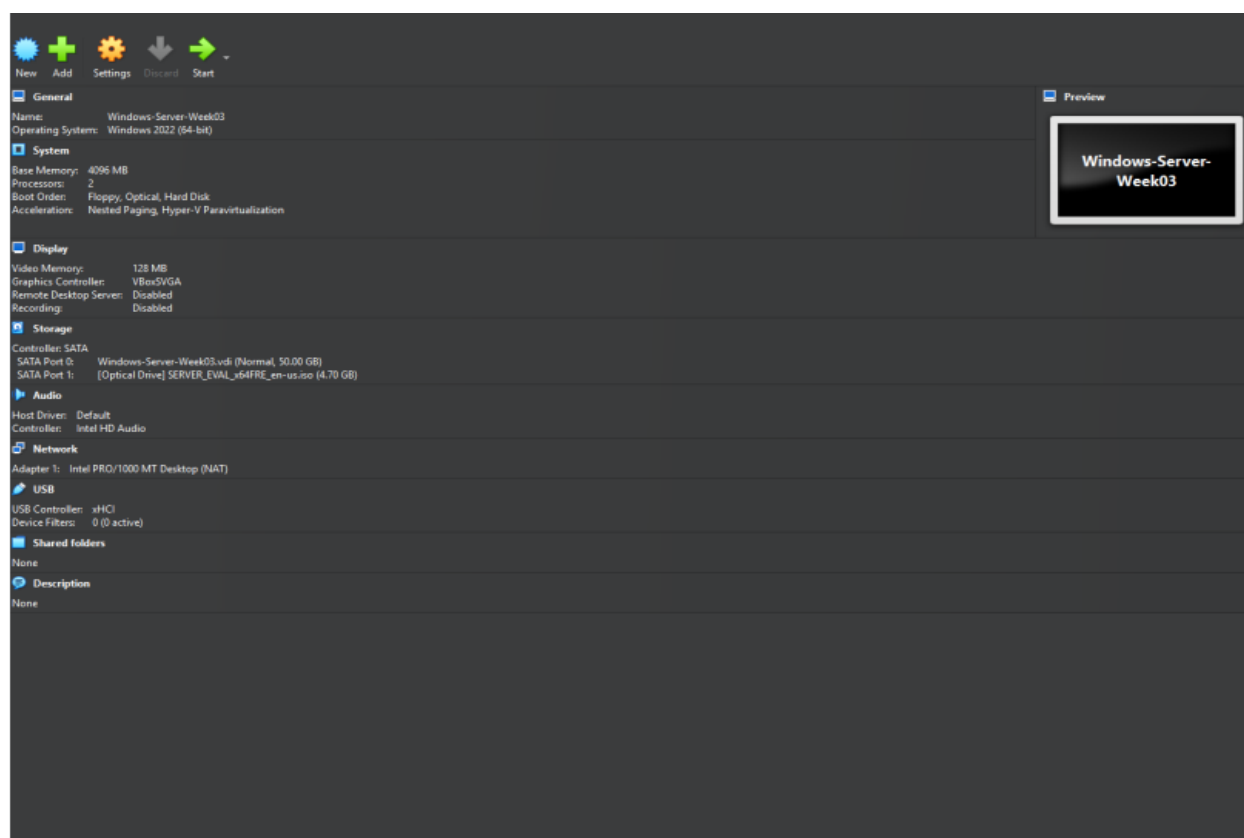


Figure 12. Oracle VirtualBox specifications for the Windows Server virtual machine.

9.2 Edition Selection

Windows Server 2022 Standard Evaluation with Desktop Experience was selected. Desktop Experience was used because it provides a graphical interface and Server Manager

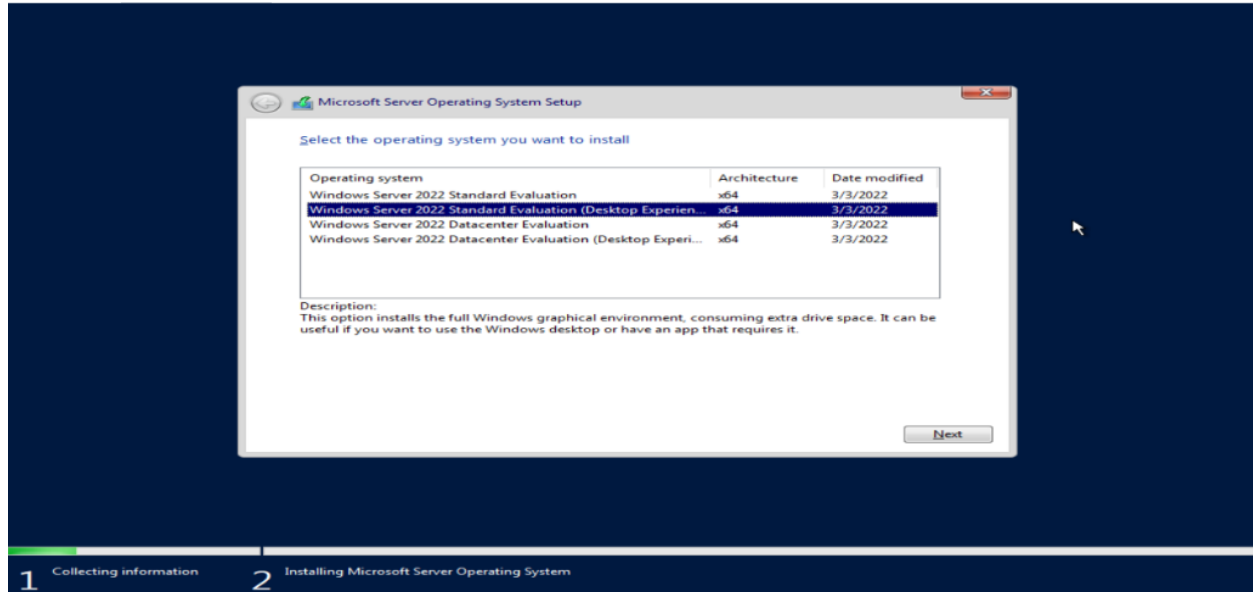


Figure 13. Selection of Windows Server 2022 with Desktop Experience.

9.3 Disk and Installation

A custom installation was performed using Drive 0 Unallocated Space. Windows Setup automatically created the necessary system partitions and installed the operating system on the virtual disk.

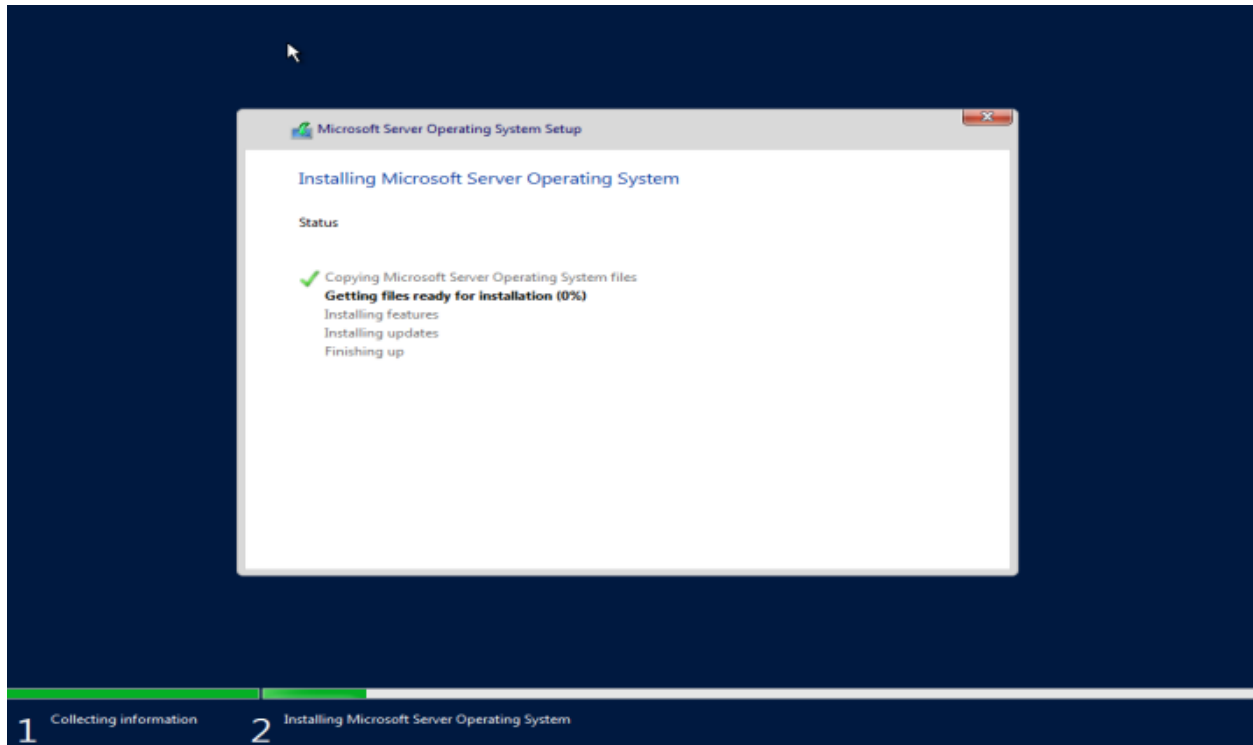


Figure 14. Windows Server installation progress.

9.4 Administrator Configuration

After installation, a password was configured for the built-in Administrator account. The password provided administrative access to Windows Server and was kept hidden in the documentation.

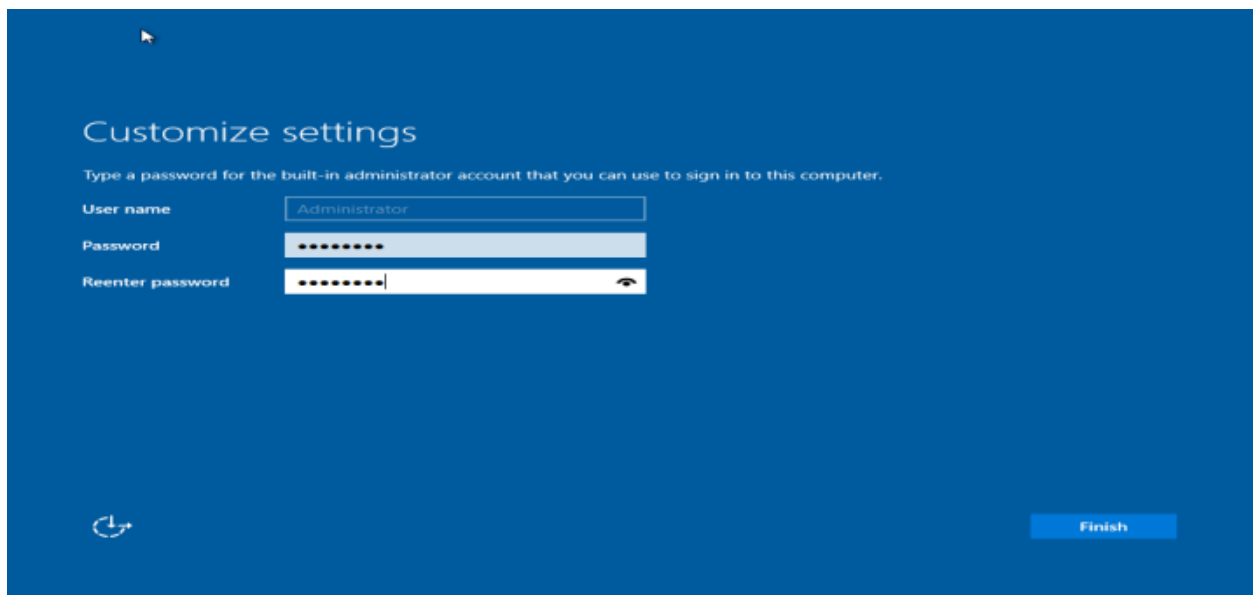


Figure 15. Administrator account password configuration.

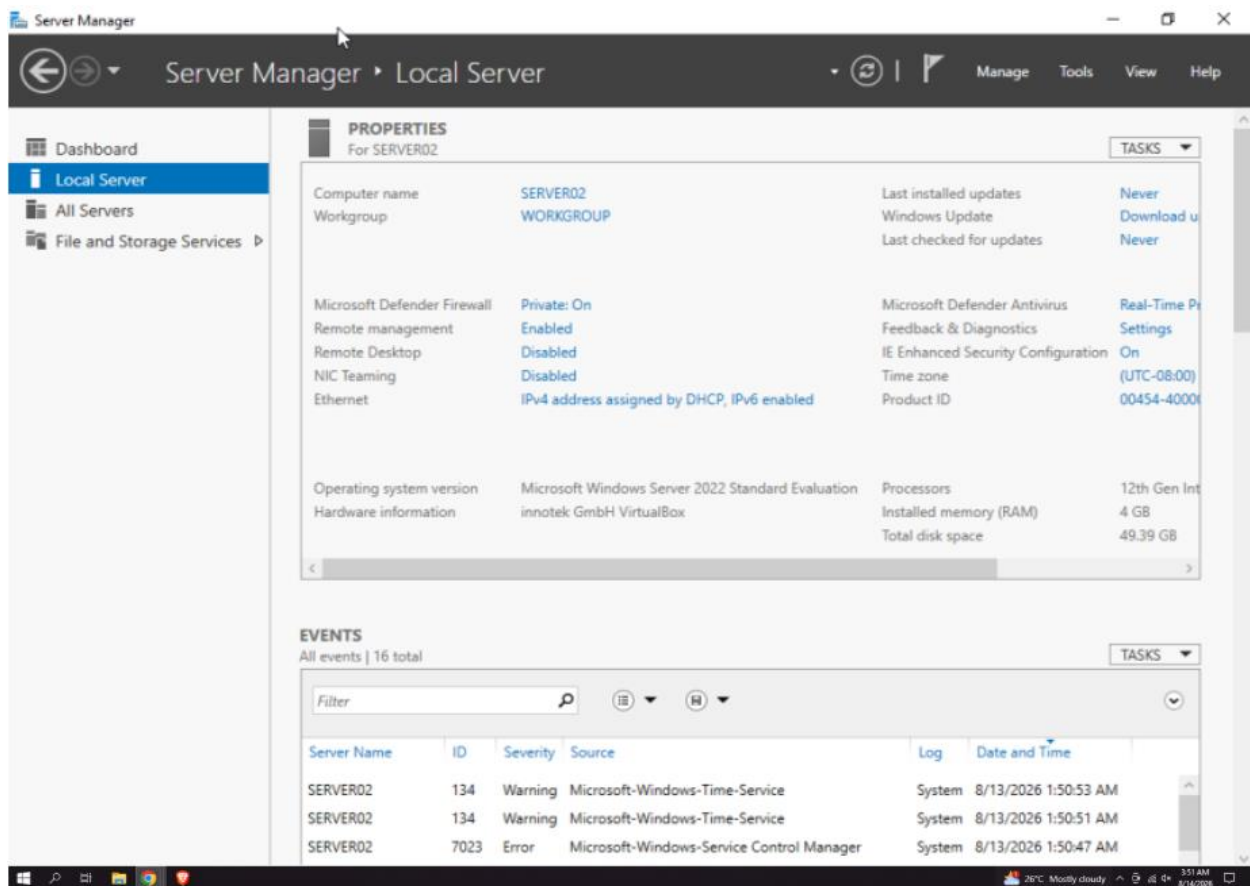


Figure 16. Successful Windows Server login and verification of the SERVER02 computer name.

10. Enterprise Operating System Comparison

Category	Windows Server 2022	Ubuntu Server 24.04.4 LTS	Rocky Linux
Licensing	Proprietary and generally requires a paid production license	Free and opensource, with optional Ubuntu Pro services.	Free and opensource, with optional third-party support.
Interface	Server Core or graphical Desktop Experience.	Primarily commandline based.	Primarily commandline based, with optional graphical environments.
Package Management	Server Manager, PowerShell, Windows Update, and installers.	APT and Ubuntu software repositories.	DNF and RPM packages.
Security	Defender, Firewall, BitLocker, Group Policy, and	AppArmor, UFW, Secure Boot, permissions, and security updates.	SELinux, firewalld, permissions, and security advisories.

	securedcore technologies.		
Performance	Strong for Microsoft enterprise workloads but may require more resources	Lightweight and efficient for cloud, web, and container workloads.	Stable and efficient for enterprise Linux workloads.
Typical Uses	Active Directory, file services, Hyper-V, and Microsoft applications.	Web hosting, databases, cloud computing, DevOps, and containers.	Enterprise applications, web hosting, databases, and Red Hatcompatible workloads
Advantages	Familiar interface and strong Microsoft integration.	Free, efficient, widely documented, and supported by a large community.	Stable, secure, free, and compatible with the Enterprise Linux ecosystem
Disadvantages	Licensing costs and comparatively higher resource requirements.	Requires commandline knowledge and may not support some proprietary applications.	Smaller community and slower availability of newer software versions.

Windows Server is a good choice for organizations that rely heavily on Microsoft Technologies and centralized domain technologies. Ubuntu Server is well-suited for cloud environments, web hosting, containerized applications, and open-source software. Rocky Linux is a suitable option for businesses that prioritize stability and compatibility with the Red Hat Enterprise Linux ecosystem.

The complete discussion is provided separately in OS_Comparison_Report.pdf.

11. Problems Encountered and Solutions Implemented

11.1 Incorrect Ubuntu ISO

The first downloaded ISO was Ubuntu Desktop rather than Ubuntu Server. Although both editions were labeled Ubuntu 24.04.4 LTS, the filename contained desktopamd64.

Solution: The correct ubuntu-24.04.4-live-server-amd64.iso file was downloaded and attached to the Ubuntu Server virtual machine.

11.2 Installation Media Could Not Be Unmounted

After Ubuntu installation, the system displayed a message stating that it failed to unmount /cdrom and requested removal of the installation media. Solution: The ISO was removed

from the VirtualBox optical drive, and the virtual machine was allowed to reboot from the installed virtual hard disk.

11.3 OpenSSH Service Was Not Found

The command used to verify SSH initially reported that `ssh.service` could not be found. This indicated that OpenSSH Server had not been installed during the installation wizard.

Solution: OpenSSH Server was installed manually using `sudo apt install opensshserver -y`. The service was then started and verified as active and running.

11.4 Incorrect Command Typing

The `systemctl` command was initially entered incorrectly, preventing the SSH service status from being displayed. Solution: The correct command syntax was reviewed. The equivalent service `ssh status` command was then used to verify the service successfully.

12. Lessons Learned

Through this activity, I gained hands-on experience installing and configuring two enterprise server operating systems. I recognized the importance of selecting the right ISO edition, as desktop and server variants from the same LTS release differ significantly in user interfaces and default packages. Additionally, I learned how adjusting virtual machine resources—such as CPU, RAM, storage, network modes, and installation media—directly impacts server performance.

Executing verification commands in Ubuntu highlighted how administrators confirm network settings, hostnames, internet connection, system updates, and active services post-installation. Troubleshooting a missing SSH service reinforced the value of precision when typing terminal commands, reading error outputs, and applying post-deployment fixes. Finally, setting up Windows Server expanded my skills in navigating GUI-based server environments, configuring Administrator credentials, and establishing proper system naming. Ultimately, this exercise strengthened my foundation in server deployment, virtualization, remote management, and documentation, preparing me for advanced tasks like web hosting, database management, and network administration.

13. References

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